

A s t r a X F r a m e w o r k

From Innovation to Industrialization

"Predictable Delivery Through Engineering Discipline and Operational Excellence"



ENGINEERING



SUPPLY CHAIN



QUALITY



MANUFACTURING



The

Delivering Class

AstraX is built to lead — not just to compete. Every decision across engineering, supply chain, quality, and manufacturing is anchored to a set of strategic objectives that define what it means to win in this market. These objectives are not aspirational statements; they are measurable commitments embedded in every milestone.



Product

Best-in-class performance, form factor, and user experience



Cost

Optimized BOM, lean design, and scalable manufacturing



Time to Market

Disciplined execution with milestone-driven development



Supply

Redundant sourcing, qualified vendors, and inventory visibility



Quality

Embedded quality gates and regulatory readiness from day one



Scalable

Production-ready designs validated through EVT, DVT, and PVT

Concept

Define vision and core requirements

Validation

Test performance and quality



Prototype

Build functional handheld units

Production

Scale manufacturing and supply

A s t r a X

"Great Products Are Built Through Repeatable Systems"

AstraX's development philosophy rests on five interconnected pillars — each essential, none optional. From rigorous engineering standards to commercial awareness embedded early in design, these pillars ensure that every team member is working within a system that produces consistent, predictable outcomes.

E n g i n e e r i n g Rigorous design, simulation, and verification practices	D o c u m e n t a t i o n Organized, searchable, and reusable knowledge base	C o m m e r c i a l BOM cost awareness and design-for-cost principles
S u p p l y Proactive sourcing and vendor qualification		E x e c u t i o n Milestone-driven cadence with full cross-functional accountability

E n g i n e e r i n g

O n e

Organizational alignment is the foundation of execution speed. The AstraX Integrated Product Team (IPT) structure ensures that every discipline — from electrical to manufacturing — operates under a unified program plan with shared objectives, shared accountability, and a direct line of sight from the Founder to the factory floor.

I P T

A single integrated team with clear reporting lines and cross-functional ownership. No silos, no handoff gaps.

- Founder → Program Leadership
- Electrical Engineering
- Mechanical Engineering
- Firmware & Software
- Supply Chain & Quality
- Manufacturing

W h y

Cross-functional alignment eliminates decision latency. When every team shares the same program plan, weekly reviews surface blockers before they become schedule risks. Outcome: faster decisions, fewer surprises, and a product that reflects the best of every discipline.



Cross-functional alignment with common objectives drives predictable delivery at every phase gate.

Deliverables

Focus

In high-performance product organizations, progress is measured by completed deliverables — not hours logged or tasks started. AstraX uses a milestone-based execution cycle that creates a clear, repeatable rhythm: Plan, Execute, Review, Approve, Release, and advance to the next milestone. Each gate is a quality checkpoint, not just a calendar marker.



Key milestone examples: Architecture Complete · Schematic Complete · PCB Layout Released · Prototype Integrated · Validation Passed

RHYTHM

P r o g r a m

A

Great programs don't run on improvisation — they run on rhythm. The AstraX Program Management OS establishes a structured weekly meeting cadence that ensures every critical decision, technical issue, and executive concern is addressed on a predictable schedule, eliminating the chaos that derails most hardware startups.

M o n d a y

Program Planning

Weekly priorities set, blockers identified, resources confirmed

W e d n e s d a y

Technical Integration

Cross-discipline engineering sync, design decisions, issue resolution

F r i d a y

Executive Review

Program status, KPI readout, risk escalation, leadership alignment



F a s t e r

No more week-long delays waiting for a meeting that isn't on the calendar



C l e a r

Every team member knows the top three program priorities at all times



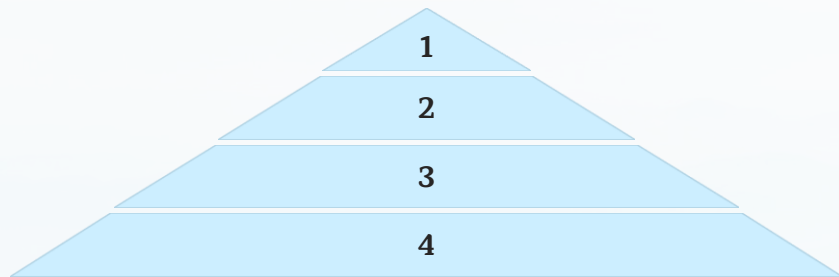
B e t t e r

Leadership has real-time awareness without micromanagement

E n g i n e e r i n g

B u i l d i n g

Documentation is not overhead — it is infrastructure. Every specification written, every design decision recorded, and every verification result captured becomes a reusable asset that accelerates future development, reduces onboarding friction, and provides the evidentiary trail required for regulatory submissions and manufacturing scale-up.



1 S y s t e m

2 A r c h i t e c t u r e

3 V e r i f i c a t i o n

4 M a n u f a c t u r i n g

D o c u m e n t a t i o n



K n o w l e d g e

Critical design rationale
survives personnel changes



F a s t e r

New team members ramp up
in days, not months



R e d u c e d

Decisions documented once
prevent repeated debates



R e g u l a t o r y
P r e p a r e d n e s s

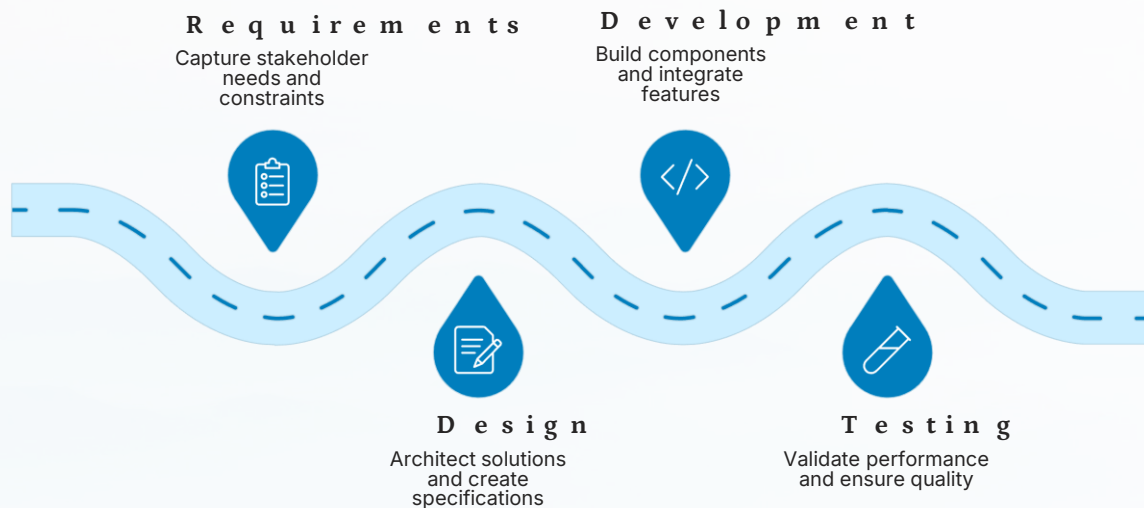
Audit-ready documentation
from the start

D i g i t a l

I n t e g r a t e d

"Knowledge Must Be Organized, Searchable and Reusable"

AstraX operates on a connected digital thread — a single, traceable chain from requirements to manufacturing support. Every tool in the ecosystem is intentionally chosen to ensure that data flows seamlessly between disciplines, reducing version conflicts, duplication, and the silent errors that emerge when teams operate in isolated silos.



G i t

Version-controlled firmware and software repositories with branching strategy

J i r a

Task tracking, sprint management, and living documentation wiki

C A D

Centralized mechanical and electrical design data with change control

D o c u m e n t

Controlled, searchable storage for all engineering and quality records

S u p p l y

B u i l d i n g - S t r a t i f i e d

Supply chain disruption is one of the top causes of hardware program delays. AstraX mitigates this risk through a structured three-tier component management model that classifies every part by strategic importance, sourcing risk, and lead time criticality — enabling the team to invest procurement energy where it matters most.

T i e r - S t r a t e g i c

Long-lead, sole-source, or custom components. Dedicated vendor relationships, buffer stock, and dual-source qualification programs.

T i e r - O p e r a t i o n a l

Semi-custom or specialty components with moderate lead times. Active monitoring, qualified alternates, and quarterly vendor reviews.

T i e r - C o m m o d i t y

Standard passives and widely available parts. Catalog sourcing with approved vendor lists and periodic cost benchmarking.



R e d u c e d

Proactive identification and mitigation of single-source dependencies



I m p r o v e d

Strategic stocking and lead-time buffering prevent production stoppages



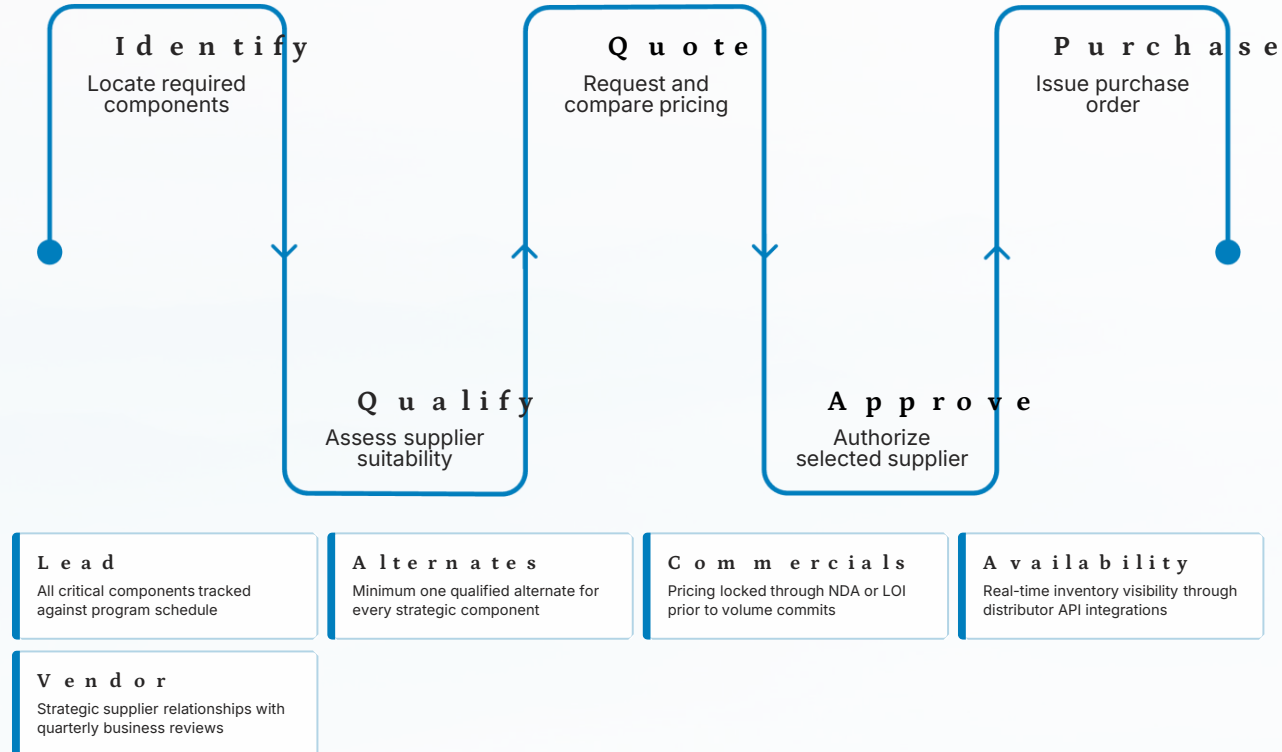
C o s t

Tiered strategy enables negotiation leverage and volume consolidation

P r o c u r e m e n t

A

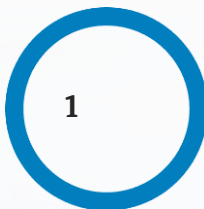
Procurement readiness is not a pre-production concern — it begins at schematic capture. By engaging the supply chain team early in the design cycle, AstraX ensures that every component on the BOM has a clear path to purchase, a qualified vendor, a confirmed lead time, and at least one validated alternate source before the first prototype build.



I n t e g r a t e d

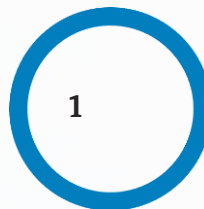
S i n g l e - R e a l t i m e

Leadership should never be surprised. The AstraX Integrated Product Dashboard consolidates all program health indicators into a single, real-time view — enabling founders, investors, and functional leads to understand program status at a glance, identify emerging risks early, and make data-driven decisions with confidence.



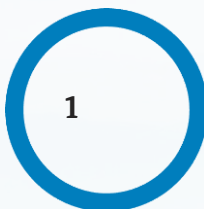
S c h e d u l e

On-time milestone tracking



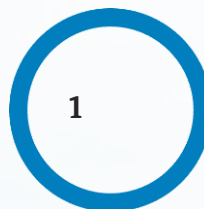
B u d g e t

Spend vs. plan visibility



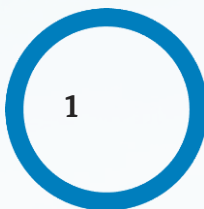
P r o c u r e m e n t

BOM coverage and vendor status



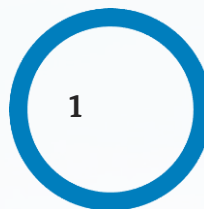
D o c u m e n t a t i o n

Completeness by phase gate



M a n u f a c t u r i n g

DFM compliance and yield projections



Q u a l i t y

Verification and validation status

Dashboard targets reflect the end-state EB1 readiness goal. Current actuals are tracked weekly and reviewed every Friday in the Executive Review cadence.



E n g i n e e r i n g

W h a t

— W h a t

Metrics are the nervous system of a high-performance product organization. AstraX tracks a carefully curated set of engineering KPIs that reflect true program health — not vanity metrics. Each metric has an owner, a target, and a weekly review cadence. The goal is objective accountability, not blame.



S c h e d u l e

Planned vs. actual milestone completion rate



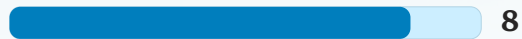
D e s i g n

Schematics, layout, mechanical, and firmware against plan



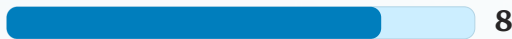
V e r i f i c a t i o n

Test cases executed vs. total required by phase gate



P r o c u r e m e n t

BOM lines with qualified vendor and confirmed lead time



D o c u m e n t a t i o n

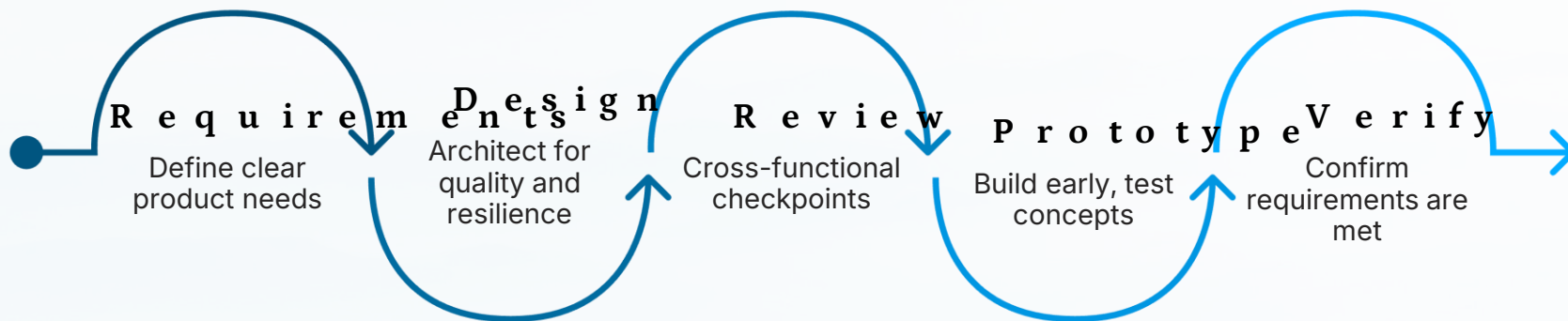
Deliverables completed vs. required for current phase

Targets shown reflect EB1 excellence benchmarks. Actuals updated weekly in the Integrated Product Dashboard.

P r o d u c t

Q u a l i t y — I t

AstraX embeds quality checkpoints at every stage of the development lifecycle. Rather than relying on end-of-cycle testing to catch failures, the quality framework ensures that requirements are testable, designs are reviewed, prototypes are systematically verified, and validation evidence is complete before any release decision is made.



R e d u c e d

Early defect detection costs 10x less than post-prototype fixes

I m p r o v e d

Systematic verification builds confidence in field performance

C u s t o m e r

First-shipment quality drives positive customer perception

M a r k e t

Compliant, validated products accelerate regulatory clearance

S u p p l y

V i s i b i l i t y

— "

The AstraX Supply Chain Command Center transforms procurement from a reactive function into a proactive strategic capability. By maintaining real-time visibility into every critical component, vendor, and purchase order, the team can anticipate shortages, accelerate at-risk items, and make informed trade-off decisions before they become program crises.



C r i t i c a l

Real-time status of all long-lead and sole-source parts with risk flags



V e n d o r

Qualification status, capacity confirmations, and NPI readiness per supplier



P u r c h a s e

PO status, confirmed delivery dates, and expedite flags for at-risk items



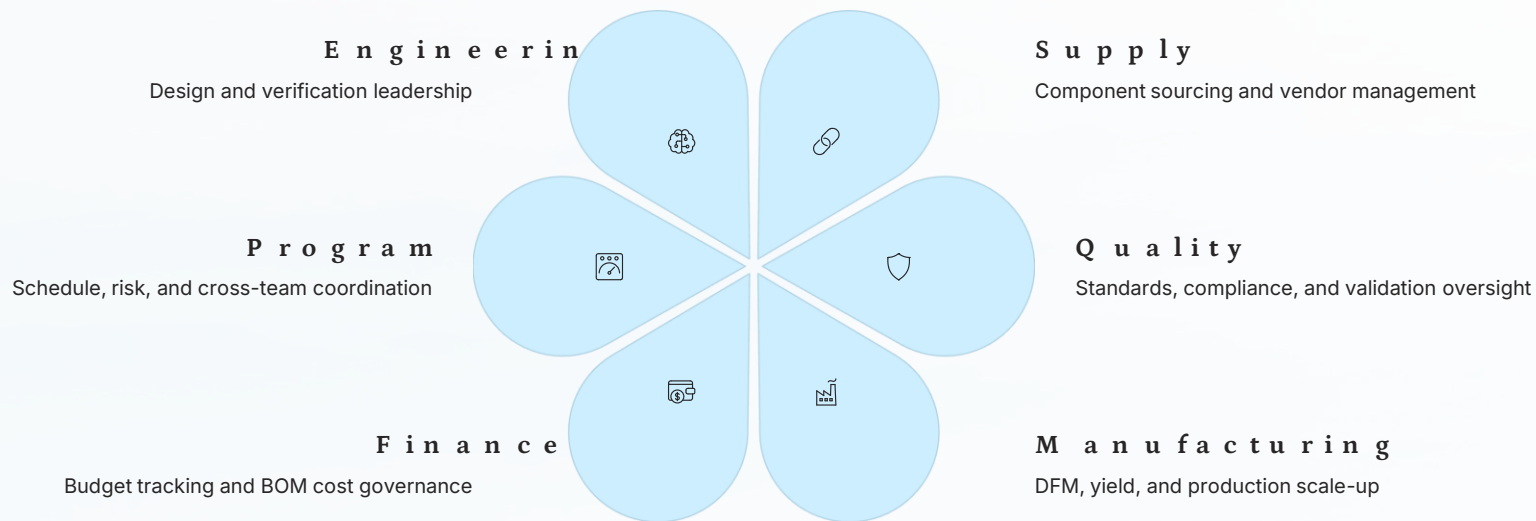
I n v e n t o r y

On-hand stock, in-transit quantities, and buffer stock targets by build phase

Cross-Functional

Every

AstraX's cross-functional model is built on the principle that silos are the enemy of speed. Engineering, supply chain, quality, manufacturing, finance, and the program office operate as an integrated team — not a sequential handoff chain. Each function has a defined contribution to AstraX customer success, and every decision is evaluated through that lens.



❏ Center of the model: **AstraX Customer Success** — every function ultimately serves the end user's experience.

M a n u f a c t u r i n g

F r o m

— A

Hardware products don't scale by accident. AstraX follows a structured manufacturing readiness journey modeled on best practices from world-class electronics OEMs. Each build phase — EVT, DVT, and PVT — has explicit entry and exit criteria, yield targets, and documented lessons learned that feed directly into the next phase's improvement plan.



D e v e l o p m e n t

Architecture finalized, DFM principles embedded from day one



P r o t o t y p e

First physical builds validate design intent and identify fit/form issues



E V T

Engineering Validation Test — functional performance and design verification



D V T

Design Validation Test — reliability, compliance, and manufacturing process validation



P V T

Production Validation Test — full production process at target yield and cost



P r o d u c t i o n

Controlled ramp to volume with continuous process monitoring and yield improvement

P r o g r a m

E n a b l i n g

Leadership in a hardware program is not about control — it is about creating the conditions for teams to move fast, make good decisions, and surface problems early. The AstraX Program Leadership Framework defines clear layers of authority, responsibility, and accountability so that every team member knows who to consult, who to escalate to, and how decisions get made.



P a r t n e r s

Execution and delivery against contracted commitments



E x e c u t i o n

Daily execution, design delivery, and issue escalation



T e c h n i c a l

Architecture decisions, trade-off resolution, cross-discipline integration



P r o g r a m

Schedule, budget, risk, and cross-functional coordination



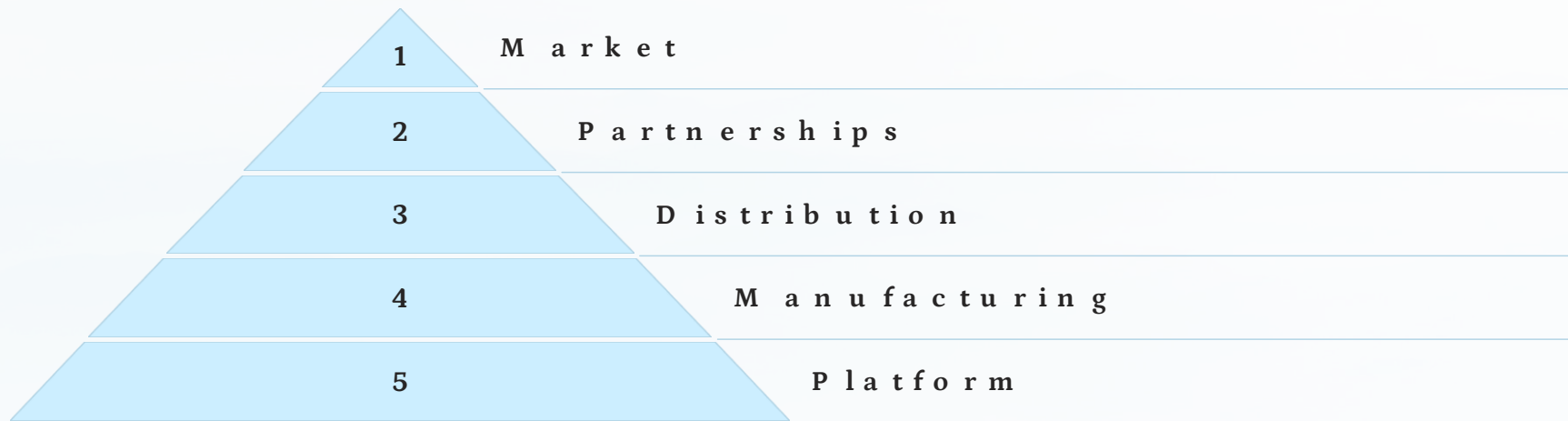
S t r a t e g i c

Vision, priorities, investment decisions, and stakeholder alignment

Scaling

Preparing — The

EB1 is not the finish line — it is the launchpad. Every engineering discipline, documentation standard, supply chain relationship, and quality process established during EB1 is designed with scale in mind. The investments made now in systems, processes, and organizational capability will compound dramatically as AstraX expands its product line, geographic reach, and manufacturing volume.



✓ EB1 Systems → EB2 Scale → Platform Expansion → Market Leadership — each phase builds on the infrastructure and discipline of the last.

A s t r a X

S i x

Predictable product delivery is not the result of one team doing exceptional work in isolation — it is the outcome of six interconnected excellence disciplines operating in harmony. When each dimension achieves its targets, the collective result is a product organization that consistently delivers on its commitments to customers, investors, and the market.

E n g i n e e r i n g

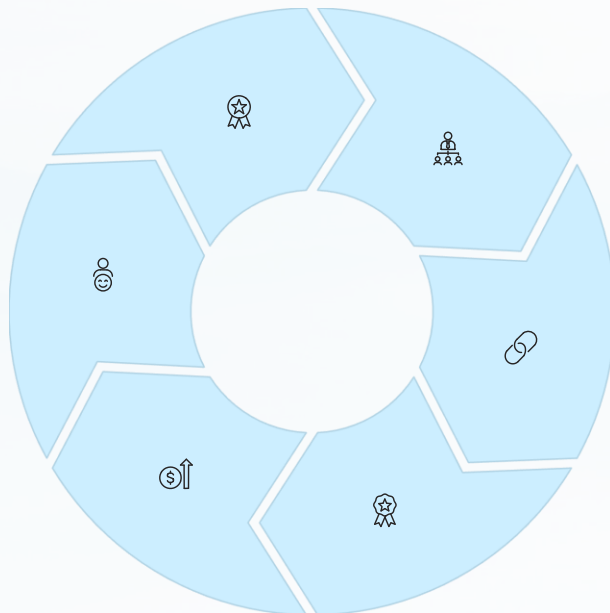
Rigorous design, simulation, and verification from requirements to release

C u s t o m e r

First-shipment quality, support readiness, and NPS-driven improvement

C o m m e r c i a l

Cost-aware design, BOM optimization, and margin discipline from day one



P r o g r a m

Milestone-driven execution with structured weekly operating cadence

S u p p l y

Resilient, risk-stratified procurement with full BOM visibility

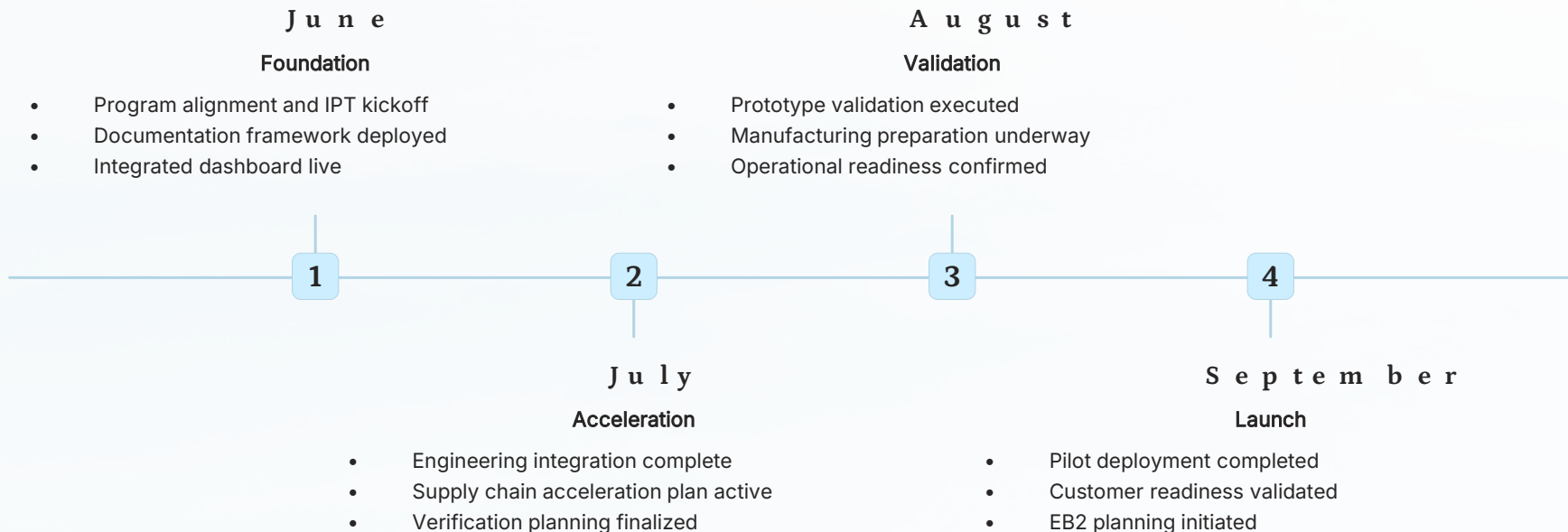
Q u a l i t y

Embedded quality gates at every phase with zero-escape mentality

9 -D a y

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The next 90 days represent the most critical execution window in AstraX's development journey. This roadmap defines clear monthly themes and deliverables, ensuring the entire organization — engineering, supply chain, quality, manufacturing — is synchronized and moving at the speed required to hit EB1 targets.



A s t r a X



P l a n

Define outcomes before committing resources



M e a s u r e

Let data drive decisions, not assumptions



I m p r o v e

Treat every phase as an opportunity to learn and improve



E x e c u t e

Honor commitments with discipline and urgency



C o l l a b o r a t e

Share information early, surface problems fast



D e l i v e r

Consistency builds trust with customers and investors

"Product Excellence Is Not an Event. It Is a Discipline Practiced Every Day."

Design → Prototype → Validation → Production → Market Success